

Roll No.: _____

Amrita Vishwa Vidyapeetham
Amrita School of Computing, Chennai
B.Tech. Mid-Term Examinations – February 2025
Second Semester
Computer Science and Engineering- Artificial Intelligence
22AIE115 User Interface Design

Time: Two hours

Maximum: 50 Marks

CO	Course Outcomes
CO01	Understand the basics of World Wide Web.
CO02	Understand the fundamentals of HTML5
CO03	Understand the fundamentals of CSS and Java Script.
CO04	Design and deploy a simple web application.

Answer all questions

1. A startup company is developing a new web-based application that requires an intuitive and user-friendly interface to enhance user experience. The development team is debating which type of user interface (UI) would be most suitable for their application. They must consider various UI types, such as Command Line Interface (CLI), Graphical User Interface (GUI), Menu-Driven Interface, Form-Based Interface, and Natural Language Interface, to determine the best fit for their platform. Analyze the different types of User Interfaces and evaluate which UI type would be most suitable for the above said web-based application. [10] [BTL4] [CO01]

2. (a) A web designer wants to create a simple portfolio showcasing their projects. The webpage should include sections like an introduction, project gallery, and contact information. Explain how basic HTML elements such as <header>, <section>, <article>, and <footer> can be used to structure a personal portfolio website. [8] [BTL 2] [CO02]

(b) Discuss how semantic HTML5 improves both the readability and accessibility of the webpage. [2] [BTL 2] [CO02]

3. Design a simple web page to display a student's grade report for a semester. The report should include the student's name, roll number, and a table displaying the subject names, marks obtained, and grades. The table should be well-structured and visually appealing. Write the code based on the following constraints.

Output Requirements:

- ✓ The webpage should display the student's name and roll number at the top.
- ✓ The table should have the following columns: Subject Name, Marks Obtained, and Grade.
- ✓ The table should include at least 5 subjects with their respective marks and grades.
- ✓ Use appropriate HTML tags to structure the content.
- ✓ Apply styles to the table (e.g., border, padding, and background colour for headers).

[10] [BTL 3] [CO02]

4. A food blog is creating a webpage for recipes. Each recipe includes a list of ingredients (unordered list) and step-by-step instructions (ordered list) for cooking the dish. Explain how HTML5 can be used to create both an unordered list for ingredients and an ordered list for step-by-step instructions in a recipe. Discuss how the tags are utilized to organize this content.

[10] [BTL4] [CO02]

5. A small business wants to create a website to showcase its products online. The owner is unsure how the web works and how users will access the website. Explain the process of how a user's browser requests a webpage from a web server and how the website is displayed.

List of Ingredients

o Brown
o Sugar

to Recipes

[10] [BTL 3] [CO01]

... 0.4 kg = 400g

Amrita School of Computing, Chennai
B.Tech. Mid-Term Examinations – February 2025

Second Semester

Computer Science and Engineering– Artificial Intelligence

22AIE112 Data Structures & Algorithms 1

Time: Two hours

Maximum:50 Marks

CO	Course Outcomes
CO01	Apply an appropriate data structure for a specified problem
CO02	Analyze the complexity of algorithms
CO03	Implement linear data structures to solve different problems
CO04	Implement non-linear data structures like trees to solve different problems.

Answer all questions

1. Write a program to perform the following operations on a Doubly Linked List that stores student details (Student No and Name) in each node:
- Traverse the list in reverse order and display the student details (Student No and Name) from the last node to the first.[4]
 - Print the total number of students present in the list.[2]
 - Insert a new student record at a specified K^{th} position in the list.[4]
- Ensure valid operations by handling the following conditions:[2]

- Display an appropriate message if the list is empty when attempting reverse traversal or counting nodes.
- Handle an invalid insertion positions ($K < 1$ or $K > \text{list size} + 1$) with an error message.
- Ensure proper memory allocation and deallocation to prevent memory leaks.

[12][BTL3] [CO03]

2. Identify a suitable data structure to evaluate the given mathematical expression:
 $((D+E*F-G)/K+L+(M*N))$, given the values as $D=1, E=2, F=3, G=4, K=6, L=6, M=7, N=8$.

[8] [BTL2] [CO01]

3. Develop a program to implement a Singly Linked List and perform the following operations:

- Remove any node from the list except the first and last node.[4]
- Search for a given element in the list and return the previous node's element if the search is successful.[4]

Ensure proper error handling by notifying the user when:[2]

- Attempt to delete the first or last node.
- Search for an element that does not exist.
- Performing any other invalid operations on the list.

[10] [BTL3] [CO03]

4. Develop a program to implement Stack operations using an appropriate implementation strategy. Perform the following operations:

- Insert an element into the stack.[2]

(ii) Remove and return the top element from the stack.[2]

(iii) Retrieve the top element without removing it.[2]

(iv) Determine whether the stack is empty or full (using a fixed-size stack).[2]

Ensure proper error handling by notifying the user when:[2]

- Attempting to pop or peek from an empty stack.
- Attempting to push an element into a full stack

[10] [BTL2] [CO03]

5. Explain the concept of a dynamic array and its advantages over a static array. Develop a program to perform a linear search on a dynamically allocated array containing 10 integer numbers. Analyze and discuss the time and space complexity of the linear search algorithm based on your implementation.

[10] [BTL3] [CO02]

Amrita Vishwa Vidyapeetham
Amrita School of Computing, Chennai
B.Tech. Mid-Term Examinations – February 2025
Second Semester

Computer Science and Engineering - AIE
22AIE113: Elements of Computing Systems - 2

Maximum: 50 Marks

Time: Two hours

CO	Course Outcomes
CO01	Analyze the important components of a MIPS computer system and the basic organization
CO02	Implement low-level programming on the hardware platform
CO03	Develop programs in object-based language 'Jack'
CO04	Execute experiments related to basic concepts and functions of operating systems and compilers.

Answer all questions

- Outline the CPU implementation for the instruction $D=D+1$;JMP with clear control flow diagram. [05] [BTL2] [CO01]
- Construct an assembly language program to compute the sum of N numbers and illustrate the conversion process into machine language. [15] [BTL3] [CO02]
 - Write the assembly code for computing the sum of N numbers. (5M)
 - Provide the step-by-step process of converting this assembly code into machine language using a two-pass assembler. (6M)
 - Use the given translation codes to represent the final machine code format. (4M)

Symbolic syntax: dest = comp ; jump

Binary syntax: 1 1 1 a c c c c c d d d j j j

comp		c	c	c	c	c	c
0		1	0	1	0	1	0
1		1	1	1	1	1	1
-1		1	1	1	0	1	0
D		0	0	1	1	0	0
A	M	1	1	0	0	0	0
!D		0	0	1	1	0	1
!A	!M	1	1	0	0	0	1
-D		0	0	1	1	1	1
-A	-M	1	1	0	0	1	1
D+1		0	1	1	1	1	1
A+1	M+1	1	1	0	1	1	1
D-1		0	0	1	1	1	0
A-1	M-1	1	1	0	0	1	0
D+A	D+M	0	0	0	0	1	0
D-A	D-M	0	1	0	0	1	1
A-D	M-D	0	0	0	1	1	1
D&A	D&M	0	0	0	0	0	0
D A	D M	0	1	0	1	0	1

a = 0 a = 1

dest	d	d	d	effect: the value is stored in
Null	0	0	0	the value is not stored
M	0	0	1	RAM[A]
D	0	1	0	D register
DM	0	1	1	D register and RAM[A]
A	1	0	0	A register
AM	1	0	1	A register and RAM[A]
AD	1	1	0	A register and D register
ADM	1	1	1	A register, D register, and RAM[A]

jump	d	d	d	effect:
Null	0	0	0	No jump
JGT	0	0	1	if comp > 0 jump
JEQ	0	1	0	if comp = 0 jump
JGE	0	1	1	if comp ≥ 0 jump
JLT	1	0	0	if comp < 0 jump
JNE	1	0	1	if comp ≠ 0 jump
JLE	1	1	0	if comp ≤ 0 jump
JMP	1	1	1	Unconditional jump

3. Develop a Jack program that demonstrates list construction, iteration, and recursive access. [15] [BTL3] [CO03]
- (a) Write a Jack program for list construction. (5M)
 - (b) Demonstrate how iteration and recursion are used for list manipulation. (5M)
 - (c) Compare iteration and recursion in terms of efficiency and explain when each approach is preferred. (5M)
4. Design a Square Dance application using Jack by applying object-oriented principles. [15] [BTL3] [CO03]
- (a) Identify the core classes involved (e.g., Dancer, Square, DanceFloor) and describe their responsibilities. (5M)
 - (b) Relate how inheritance, encapsulation, and polymorphism are applied in the design. (5M)
 - (c) Apply these principles to demonstrate how they contribute to modularity and reusability in the application. (5M)

Amrita School of Computing, Chennai
B.Tech. Midterm Examinations – February 2025
Second Semester
Artificial Intelligence

22MAT122: MATHEMATICS FOR COMPUTING 2

Time: Two hours

Maximum: 50 Marks

COs	Course Outcomes
CO1	Apply matrix decomposition techniques to solve elementary problems.
CO2	Apply the concepts of linear algebra and differential calculus to solve elementary optimization problems
CO3	Analyze data using fundamental techniques of probability.
CO4	Implement the concepts and techniques of linear algebra, optimization and probability for signal and image processing.

Answer all questions

1. Solve the system of equations using LU decomposition method

$$\begin{aligned}x + y + z &= 1 \\4x + 3y - z &= 6 \\3x + 5y + 3z &= 4\end{aligned}$$

[5] [BTL 3] [CO1]

2. (a) Find the Fourier Transform of $f(x) = \begin{cases} 1 - x^2, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$. Hence deduce that

$$\int_0^\infty \left(\frac{\sin t - t \cos t}{t^3} \right)^2 dt = \frac{\pi}{15}.$$

[5] [BTL 3] [CO1]

- (b) Using Cayley Hamilton theorem find A^{-1} for $A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$. [5] [BTL 3] [CO1]

3. Diagonalise the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ -4 & 4 & 3 \end{bmatrix}.$$

[5] [BTL 4] [CO1]

4. (a) Find Singular Value Decomposition of $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}$.

[6] [BTL 2] [CO2]

- (b) Find the Fourier series of $f(x) = \begin{cases} x, & 0 < x < 1, \\ 1 - x, & 1 < x < 2. \end{cases}$

[4] [BTL 2] [CO2]

5. State and prove change of scale property for Fourier transform.

[5] [BTL 2] [CO1]

6. Use Taylor's theorem to expand $f(x, y) = e^y \ln(1 + x)$ in powers of x & y .

[5] [BTL 3] [CO2]

7. (a) The temperature T at any point (x, y, z) in space is $T(x, y, z) = 100xyz^2$. Find the highest temperature on the surface of the sphere $x^2 + y^2 + z^2 = 1$.

[7] [BTL4] [CO2]

(b) Examine whether the vectors $V = \{(1, 1), (4, 4)\}$ are linearly independent or dependent.

[3] [BTL4] [CO2]

Amrita Vishwa Vidyapeetham
Amrita School of Computing, Chennai
B.Tech. End Semester Examinations – April/May 2025
Second Semester
Computer Science and Engineering– Artificial Intelligence
22AIE112 Data Structures & Algorithms 1

Time: Three hours

Maximum:100 Marks

CO	Course Outcomes
CO01	Apply an appropriate data structure for a specified problem
CO02	Analyze the complexity of algorithms
CO03	Implement linear data structures to solve different problems
CO04	Implement non-linear data structures like trees to solve different problems.

Answer all questions

- Give a detailed analysis of Big-Oh of running time for the code snippet:
`sum=0;`
`for(i = 0; i < n; ++i )`
`++sum;`
[2][BTL2] [CO02]
- List the difference between linear and non-linear data structure. [2][BTL2][CO01]
- Construct an expression tree for the equation $(a+b)/(c-d)$. [2][BTL2][CO04]
- Show that in a binary tree of N nodes, there are $N + 1$ nullptr links representing children. [2][BTL2][CO04]
- Can insert and findMin be implemented in constant time in minheap? Justify your answer. [2][BTL3][CO02]

yes.
- List1 and List2 are two sorted linear ListADT. Develop a program to create a sorted List where elements from List1 and List2 are merged. [10][BTL3] [CO03]

Test Case 1:

Input:

List1 = [1, 3, 5]

List2 = [2, 4, 6]

Output: [1, 2, 3, 4, 5, 6]

Test Case 2:

Input:

List1 = []

List2 = [2, 4, 6]

Output:[2, 4, 6]
- Develop a program to check for balancing symbols. Find a suitable data structure to implement the balancing symbols. (`/* */`, `()`, `[]`, `{ }`). [10][BTL3][CO01]

Test Case 1:

Input

```
if(a>b)
```

```
{printf("a is greater");}
```

Output: Balanced

Test Case 2:

Input

```
/* if condition
```

```
if(true)
```

```
{printf("condition satisfied");}
```

Output: Not Balanced

8. Show the result of the following sequence of instructions: union(1,2), union(3,4), union(3,5), union(1,7), union(3,6), union(8,9), union(1,8), union(3,10), union(3,11), union(3,12). performing union-by-height. [10][BTL3][CO04]
9. (a) Demonstrate the result of inserting the key 10, 12, 1, 14, 6, 5, 8, 15, 3, 9, 7, 4, 11, 13, and 2, one at a time, into an initially empty binary heap [10M]
(b) Show the result of performing three deleteMin operations in the heap in the constructed heap [5M].
(c) Write the routine for insert and deletemin. [5M] [20][BTL3][CO04]
10. A deque is a data structure consisting of a list of items on which the following operations are possible:
(a) **push(x)**: Insert item x on the front end of the deque. [5M]
(b) **pop()**: Remove the front item from the deque and return it. [5M]
(c) **inject(x)**: Insert item x on the rear end of the deque. [5M]
(d) **eject()**: Remove the rear item from the deque and return it. [5M]
Write routines to support the deque that take $O(1)$ time per operation. [20][BTL3][CO03]
11. (a) Show the result of inserting the key 66, 43, 39, 86, 6, 14, 2, 77, 45, 30 into an initially empty AVL Tree. [10M]
(b) Write the insert procedure. [5M]
(c) Perform Preorder and Postorder traversal of the constructed AVL tree. [5M] [20][BTL3][CO04]

Roll No.: _____

Amrita Vishwa Vidyapeetham
Amrita School of Computing, Chennai
B.Tech. End Semester Examinations – April/May 2025
Second Semester
Computer Science and Engineering- Artificial Intelligence
22AIE115 User Interface Design

Time: Three hours

Maximum: 100 Marks

CO	Course Outcomes
CO01	Understand the basics of World Wide Web.
CO02	Understand the fundamentals of HTML5
CO03	Understand the fundamentals of CSS and Java Script.
CO04	Design and deploy a simple web application.

Answer all questions

1. A digital agency is developing a portfolio website. They implement CSS Grid for the layout but notice unexpected gaps and misaligned elements. Some sections collapse when viewed on smaller screens. Analyze the possible misconfigurations in the CSS Grid layout. How would you modify the CSS rules to fix alignment issues while maintaining responsiveness.

[10] [BTL3] [CO03]

2. Write CSS code to create a spinning loader animation based on the given constraints

- The spinner should rotate continuously.
- It should use CSS animations instead of JavaScript.
- The animation should pause when hovered over.

@keyframes
transform (400px)
transform (100%)
[10] [BTL4] [CO03]

3. A corporate website needs a unified look across multiple pages. The design team wants to use external CSS to ensure consistency in fonts, colors, and layout, while making the site easier to maintain.

(A) Explain how external CSS can be used to style a corporate website to ensure a consistent look across multiple page. [6]

(B) Discuss the advantages of using external CSS over inline or embedded styles in large web applications [4]

Conce

[10] [BTL3] [CO03]

4. A web designer wants to create a simple portfolio showcasing their projects. The webpage should include sections like an introduction, project gallery, and contact information. Explain how basic HTML elements such as `<header>`, `<section>`, `<article>`, and `<footer>` can be used to structure a personal portfolio website. Discuss how semantic HTML improves both the readability and accessibility of the webpage [10] [BTL2][CO01]

5. A freelance web developer is customizing a landing page for a client. The client wants a unique style applied to only one specific section of the page—like a featured product banner. The developer decides to use a CSS selector that targets a single, unique element without affecting others. As the developer, identify the CSS selector defined using the symbol `#`. Justify its use by providing a suitable example program that demonstrates how it applies styling to a uniquely identified HTML element. [10] [BTL3][CO03]

6. A client wants to create a secure form for user registration with these requirements:

- o First name and last name should not be numeric.
- o Email field must match the pattern `username@domain.com`.
- o Passwords must be between 8 and 15 characters and match the confirmation password.

Write a complete form and corresponding JavaScript to validate the above conditions. Display appropriate error messages using `<span>` elements near each input field. [10] [BTL3] [CO03]

7. Design a web application to find an element by its element id use the required DOM method for the following and also construct the DOM tree for the same. [10] [BTL3] [CO04]

8. Create a JavaScript program that simulates a simple game using a do-while loop, where the user has to guess a random number within a specified range. The loop should continue until the user guesses the correct number [10] [BTL4] [CO03]

9. Write a JavaScript program that uses a try-catch-finally block to handle exceptions. Explain the purpose of the 'finally' block and how it differs from the 'catch' block [10] [BTL4] [CO03]

10. SafeCart is an e-commerce startup. During usability testing, users accidentally clicked the "Place Order" button without reviewing their cart. The team decided to introduce a warning system that reminds users to double-check their cart before confirming the order. As a JavaScript developer, you are assigned to enhance the checkout page using alert dialogs.

Amrita Vishwa Vidyapeetham
Amrita School of Computing, Chennai
B.Tech. End Semester Examinations – April 2025
Second Semester
Artificial Intelligence Engineering

22MAT122: MATHEMATICS FOR COMPUTING 2

Time: Three hours

Maximum: 100 Marks

COs	Course Outcomes
CO1	Apply matrix decomposition techniques to solve elementary problems.
CO2	Apply the concepts of linear algebra and differential calculus to solve elementary optimization problems
CO3	Analyze data using fundamental techniques of probability.
CO4	Implement the concepts and techniques of linear algebra, optimization and probability for signal and image processing.

Answer all questions

1. Find Singular Value Decomposition of $A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$. [10] [BTL 3] [CO1]
2. Using the Fourier Transforms evaluate $\int_0^\infty \frac{1}{(m^2+49)(m^2+36)} dm$. [7] [BTL 2] [CO2]
3. Diagonalize $A = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$. [7] [BTL 3][CO1]
4. Find the Fourier series of $f(x) = \begin{cases} \pi x, & 0 \leq x \leq 1, \\ \pi(2-x), & 1 \leq x \leq 2. \end{cases}$ Hence deduce that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$. [8] [BTL 3] [CO2]
5. Use Taylor's theorem to expand $f(x,y) = e^x \ln(1+y)$ in powers of x & y . [7] [BTL 2] [CO2]
6. Perform two iterations of the Newton's method to minimize the function $f(x,y) = 100(y-x^2)^2 + (1-x)^2$ with $X^{(1)} = \begin{pmatrix} -1.2 \\ 2.5 \end{pmatrix}$. [10] [BTL4] [CO2]

7. Find the multivariate regression for the given data,

Handwritten notes and calculations:

$$\int_0^\infty \frac{1}{(m^2+7^2)(m^2+6^2)} dm$$

Handwritten signature: *Abhishek*

X_0	0	1	2	3	4
Y_1	1	4	3	8	9
Y_2	-1	-1	2	3	2

[10] [BTL4] [CO3]

8. Suppose $P(x) = \begin{cases} \frac{(e^{-2} 2^x)}{x!} & \text{for } x = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$. Obtain the cumulant generating function of X . Show that all cumulants have the same value 2. Hence, deduce the mean and variance. Also, obtain the r^{th} cumulant of $Y = \frac{(X-7)}{5}$. [7] [BTL3] [CO4]

9. The number of telephone calls received by an office on average is 4 per minute. Find the probability that:
a) No call will arrive in a given one-minute period. $\lambda = 4$
b) At least two calls will arrive in a given one-minute period. $\lambda = 4$
c) At least three calls will arrive in a given two-minute period. $\lambda = 8$ [2+2+4] [BTL3] [CO3]

10. The peak temperature T , as measured in degrees Fahrenheit, on a particular day is the Gaussian (85, 100) random variable. Find the following probabilities
(a) $P(T > 100)$ (b) $P(T < 60)$ and (c) $P(70 \leq T \leq 100)$? [4+3+3] [BTL2] [CO3]

11. The joint p.d.f. of X and Y is given by $f(x, y) = e^{-(x+y)}$, $x > 0, y > 0$, find the probability density function of $U = \frac{X+Y}{2}$ with $V = Y$. [10] [BTL4] [CO4]

12. During a rainy season on a tropical island the lengths of a shower have an exponential distribution with the parameter $\theta = 2$, time being measured in minutes.
(i) What is the probability that a shower will last for more than 2 minutes?
(ii) If a shower has already lasted for 2 minutes, what is the probability that it will last for at least 3 more minutes?

$$P(X > 3) / P(X = 2)$$

[3+3] [BTL2] [CO3]

Amrita School of Computing, Chennai
B.Tech End Semester Examinations – April/May 2025
Second Semester

Computer Science and Engineering– Artificial Intelligence

22AIE111- OBJECT ORIENTED PROGRAMMING IN JAVA

Time: Three hours

Maximum: 100 Marks

CO	Definition
CO1	Represent the problems using objects and classes.
CO2	Implement object-oriented concepts using the Java language.
CO3	Apply the object-oriented concepts to design and visualize programs using UML.
CO4	Implement applications using object-oriented features.

Answer all questions

1. Explain the following concepts in Java with appropriate examples. [15] [BTL2] [CO2]
 - (i) Constructor and its types [4]
 - (ii) Static member and static method [4]
 - (iii) Packages [4]
 - (iv) Arrays [3]
2. Define a class Worker and derive classes Daily Worker and Salaried Worker from it. Every worker has a name and a salary rate. Write a method ComPay(hours) to compute the weekly pay of every worker. A Daily Worker is paid on the basis of the number of days he/she works. The Salaried Worker gets paid with the wage for 40 hours a week no matter what actual hours are. Test this program to calculate the pay of workers. Identify the required concepts and implement the same. [15] [BTL3] [CO1]
3. Write a program in Java to calculate the monthly electricity bill of a consumer according to the units consumed. The tariff is given below:

Units Consumed	Charge
Upto 100 units	Rs.1.25 per unit
For next 100 units	Rs.1.50 per unit
More than 200 units	Rs.1.80 per unit

Unit consumed = Present reading – Previous reading
Use a function named calc(int u) and print the information in the main function. [15] [BTL3] [CO4]
4. Write a Java program to create a student examination database system that prints the mark sheet of students. Input student name, marks in 6 subjects. This mark should be between 0 and 100.

If the average of marks is ≥ 80 then prints Grade 'A'.

If the average is < 80 and ≥ 60 then prints Grade 'B'.

If the average is < 60 and ≥ 40 then prints Grade 'C'.

else prints Grade 'D'.

[10] [BTL3] [CO2]

5. Define a calculator class with required methods and create an object to execute the object of calculator using the given interfaces.

```
interface Add_Sub {  
    public void add(double x, double y);  
    public void subtract(double x, double y);  
}
```

```
interface Mul_Div {  
    public void multiply(double x, double y);  
    public void divide(double x, double y);  
}
```

```
interface Calculator extends Add_Sub, Mul_Div {  
    public void printResult(double result);  
}
```

[10] [BTL3] [CO2]

6. Create software for departmental stores to maintain the following details like item_no, item description, requested quantity, cost price. Provide the options to update the stock. Calculate the selling price ($SP = CP + 20\%$). Create an exception whenever the selling price of an item exceeds the given amount. [10] [BTL3] [CO3]

7. Write a Java program to Read data from file and to write data to file. [10] [BTL3] [CO4]

8. Sketch the life cycle of multithreading with its stages. [8] [BTL3] [CO4]

9. Analyze and discuss the two approaches for creating threads using an example [7] [BTL3] [CO4]

Amrita Vishwa Vidyapeetham
Amrita School of computing, Chennai
B.Tech. Mid-Term Examination: 2024-2025
Computer Science and Engineering (Artificial Intelligence)

22AIE114 Introduction to Electrical and Electronics Engineering

Time: Two hours

Maximum: 100 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO1	Familiarize the fundamental concepts in electrical and electronics engineering.
CO2	Implement the state-of-the-art computational techniques that can be employed to analyse the structured problems in electrical engineering.
CO3	Realize basic electronic components and circuits using various semiconductor devices
CO4	Implement various circuits applications in the perspective of electronics

Answer all questions

Part A – Basic electric circuits – 15 marks - CO1

1. A residential home has the following electrical appliances:

A residential home has two lamps, each with a power rating of 85 watts (W), which are operated for 6 hours per day. Additionally, the home has a fan with a power rating of 80 watts (W), operated for 8 hours daily, a refrigerator with a power rating of 250 watts (W) that runs 24 hours per day, and a TV with a power rating of 100 watts (W), used for 3 hours each day. The energy meter in the home provides readings in kilowatt-hours (kWh).

- Calculate the total daily energy consumption (in kWh) for all the appliances combined.
- If the cost of electricity is Rs.6.0 per kWh, calculate the total daily electricity cost.
- Determine the monthly cost of electricity if the usage pattern remains the same every day.
- If the rate per kWh increases by 10%, calculate the new daily cost.
- How much energy (in kWh) would be saved if one lamp is replaced by an LED lamp with a power rating of 20 watts, assuming the same operating hours?

[10] [BTL 3] [CO1]

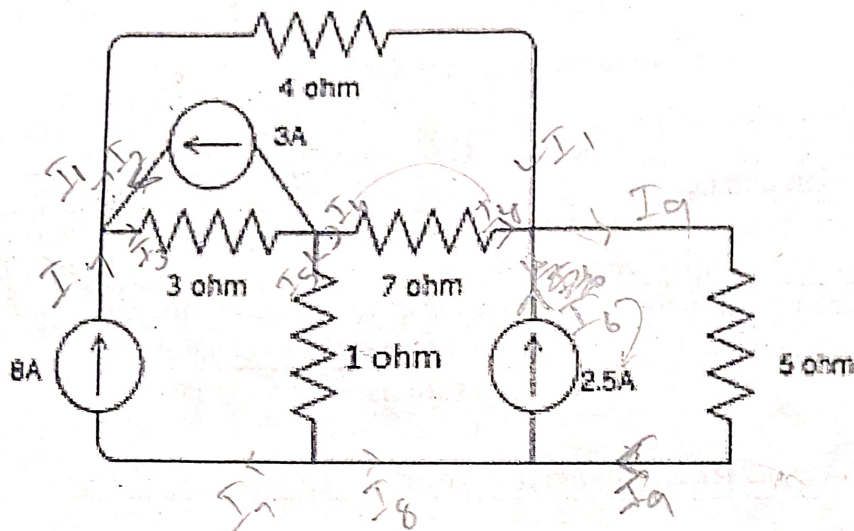
2. Determine the total charge entering a terminal between $t = 1$ sec and $t = 8$ sec, if the current

passing the terminal is $i = (3t^2 - t)$ A.

Part B – Analyze the DC circuits – 35 marks – CO2

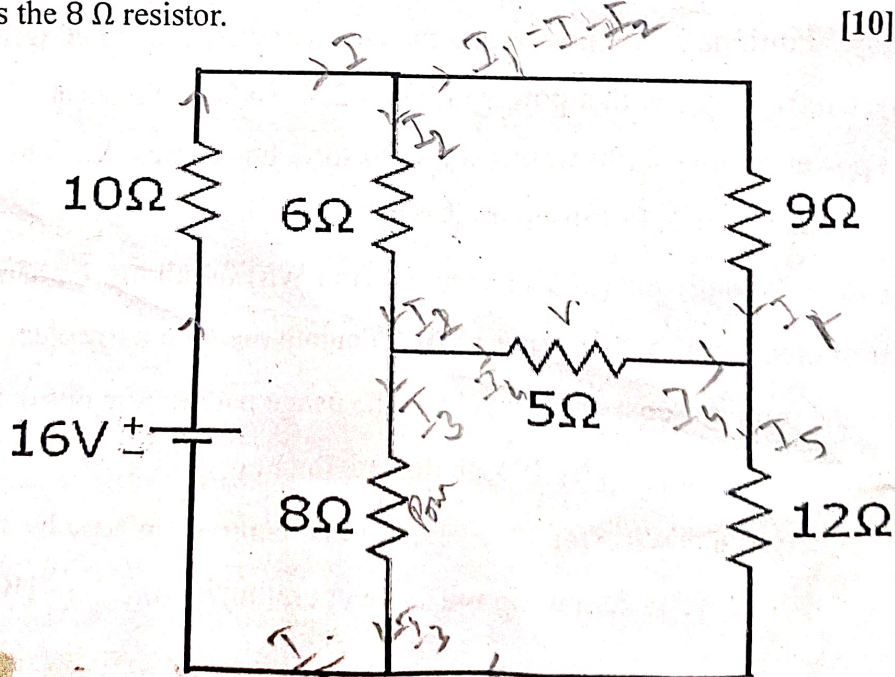
3. Solve the following network using nodal analysis and find the all-branch currents.
All resistors are in ohms.

[10] [BTL3] [CO2]

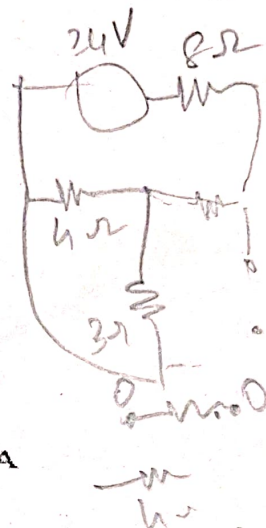
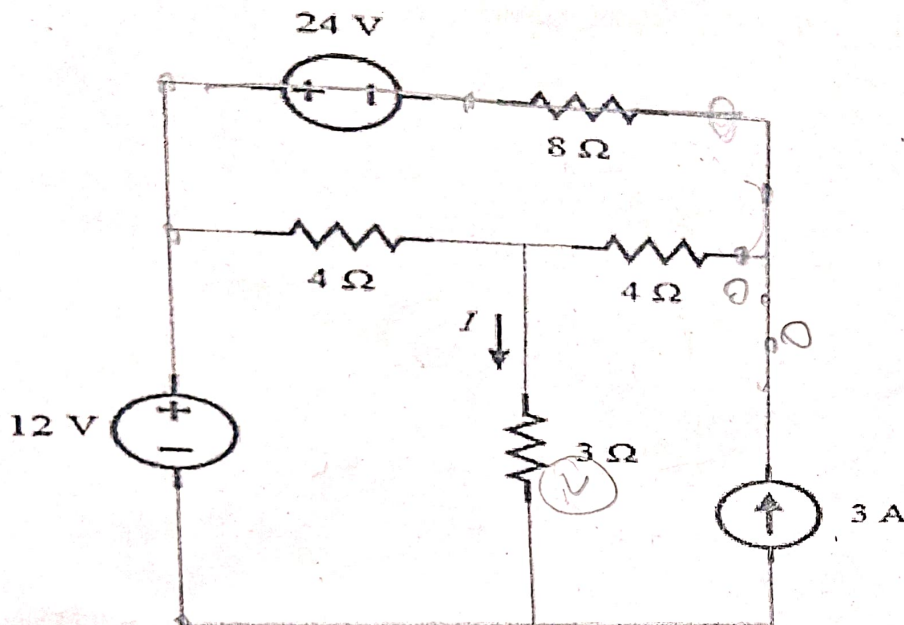


4. Solve the following network using mesh analysis and find the voltage 5Ω resistor and power absorbed across the 8Ω resistor.

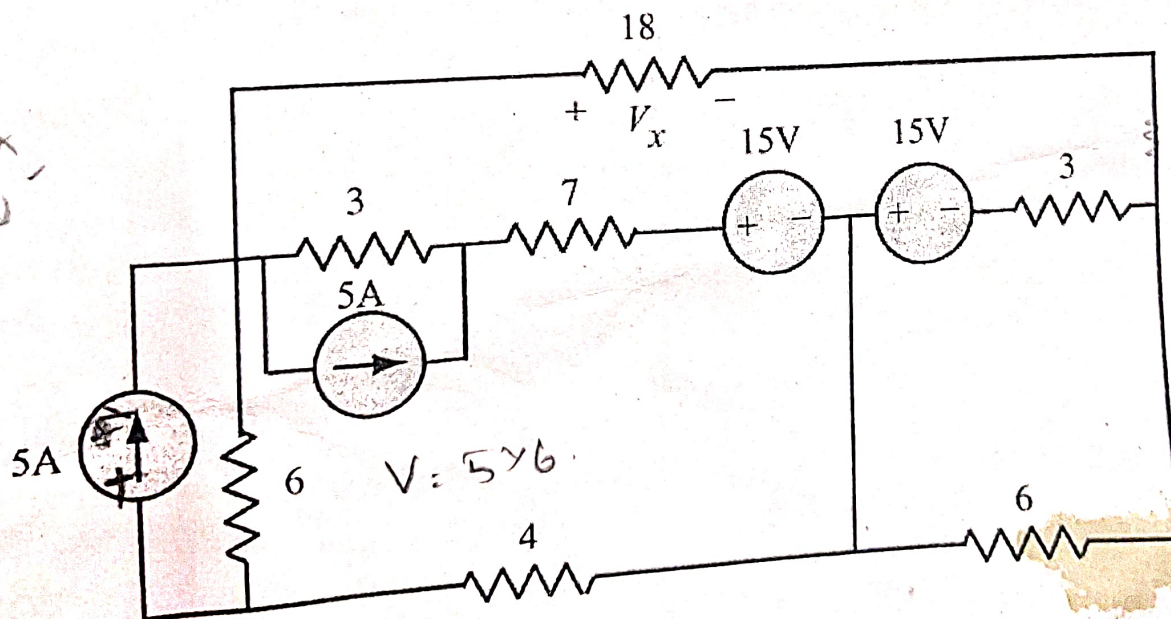
[10] [BTL 3][CO2]

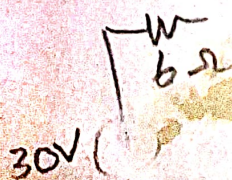


5. Apply the superposition theorem for the below network and find the total voltage across the $3\ \Omega$ resistor. Mention which voltage source provides the dominant current to the $3\ \Omega$ resistor. [10] [BTL 3] [CO2]



4. Use the source transfer technique to reduce the network shown in Fig. 1.36(a) and find the value of V_x . All resistors are in ohms. [5][BTL][CO2]





Amrita Vishwa Vidyapeetham
Amrita School of Computing, Chennai
B.Tech. End Semester Examinations – April 2025
Second Semester

Computer Science and Engineering - AIE

22AIE113: Elements of Computing Systems - 2

Time: Three hours

Maximum: 100 Marks

CO	Course Outcomes
CO01	Analyze the important components of a MIPS computer system and the basic organization
CO02	Implement low-level programming on the hardware platform
CO03	Develop programs in object-based language 'Jack'
CO04	Execute experiments related to basic concepts and functions of operating systems and compilers.

Answer all questions

1. Analyze the importance of Program Counter (PC) in the Hack CPU, explaining its role in managing instruction execution and ensuring sequential control flow. [10] [BTL2] [CO01]
2. Jack is compiled into an intermediate Virtual Machine (VM) code before being translated into Hack machine language. This process involves multiple stages. Explain the two-stage compilation process of Jack programs, covering the role of the tokenizer, parser, and code generation. [10] [BTL2] [CO03]
3. The Jack language supports recursive function calls, allowing for elegant solutions to problems like computing the Fibonacci sequence. Write a Jack program that defines a class Fibonacci containing a recursive function compute which calculates the nth Fibonacci number. [10] [BTL3] [CO04]
4. Write a simple stack data structure in Jack language with push and pop operations and explain how memory management is handled in your implementation. [10] [BTL3] [CO03]
5. Hack machine language is a 16-bit instruction set designed for a simple yet powerful computational model. It consists of A-instructions for memory addressing and C-instructions for computation and control flow. [15] [BTL3] [CO02]
(a) Describe the step-by-step process of how an assembler translates hack assembly instructions into machine language and its significance of symbol resolution and instruction encoding in this process. (10M)

- (b) Translate the following Hack assembly code into its machine language and explain each step of the conversion- (5M)

```
@2
D=A
@3
D=D-A
@4
M=D
```

6. A programmer is implementing Binary Search in Jack to efficiently search for a key in a sorted array. However, the function below may contain errors and inefficiencies. [15] [BTL3] [CO03]
Consider the following Jack implementation:

```
class BinarySearch {
  function int search(int key, Array arr, int low, int high)
  {
    var int mid;
    while (low <= high) {
      let mid = (low + high) / 2;
      if (arr[mid] = key) {
        return mid;
      } else if (arr[mid] < key) {
        let low = mid + 1;
      } else {
        let high = mid - 1;
      }
    }
    return -1;
  }
}
```

- a) Identify the logical or syntax errors in the function and explain how they affect the execution. (5M)
- b) Rewrite this function with necessary corrections to ensure a proper Binary Search implementation. (5M)
- c) Analyze how Jack's limited features impact recursion and iterative approaches in algorithm design. (5M)
7. In low-level programming, counters are essential for iteration and control flow. The Hack assembly language provides basic arithmetic and jump instructions to implement such operations. [15] [BTL3] [CO02]
- (a) Write a Hack assembly program that counts from 1 to 10, storing each number sequentially in RAM [0]. (10M)
- (b) Modify the program to count in reverse, from 10 to 1, and explain the necessary changes.

(5M)

8. Efficient handling of user input and output is crucial for interactive programs. Jack OS provides various functions such as `readChar()`, `readLine()`, and `printString()` to facilitate these operations.

[15] [BTL3] [CO04]

(a) Differentiate between `readChar()` and `readLine()` and in which scenario would each function be preferred? (5M)

(b) Develop a Jack OS-based logic that continuously accepts user input and echoes it back until the user presses the Enter key. (5M)

(c) Evaluate how the `printInt()` function processes and displays both positive and negative numbers in Jack OS. (5M)